

NAVLE Review

Exotics & Avian Study Guide

A complete species-by-species review of exotic, avian, aquatic, and small-mammal medicine.

Avian — Psittacines & pet birds · Poultry, turkeys & gamebirds

Aquatic — Ornamental fish & aquaculture

Herptiles — Reptiles & amphibians

Small mammals — Rabbits · Ferrets · Guinea pigs · Chinchillas · Hamsters, rats & mice · Hedgehogs

Primates — Nonhuman primate medicine

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HOW TO USE THIS GUIDE

Each chapter opens with the highest-yield differentiating features, then moves through organized disease tables. **Yellow boxes** are board pearls and buzzword-to-diagnosis shortcuts. **Blue boxes** are look-alike discriminators — the "how do I tell these two apart" questions the NAVLE loves. **Red boxes** flag reportable and zoonotic diseases and genuine clinical dangers.

R = reportable **Z** = zoonotic

1 | Pet Birds & Psittacines

THE THREE QUESTIONS THE NAVLE ASKS OVER AND OVER

1. **Polyomavirus vs. PBFD (circovirus)** — acute nestling death vs. chronic feather loss.
2. **Chlamydia psittaci** — zoonotic + reportable, biliverdinuria, doxycycline.
3. **Hypovitaminosis A** — all-seed diet, squamous metaplasia, white plaques, sinusitis.

1.1 Viral Disease

Disease	Agent	Signs / Lesions	Diagnosis & Management
Psittacine Beak & Feather Disease (PBFD)	Circovirus	Progressive, symmetric feather loss and abnormal feather regrowth over months; abnormal beak growth and beak lesions; immunosuppression with recurrent secondary infections. Young psittacines. Botryoid basophilic INTRACYTOPLASMIC inclusions in feather follicle epithelium and macrophages.	PCR ; quarantine affected/carrier birds. Transmitted by fomites (feather dust is highly environmentally stable). No treatment.
Budgerigar Fledgling Disease	Polyomavirus	Peracute death in nestlings/fledglings. Abdominal distention, SQ hemorrhage , delayed crop emptying / crop stasis, ascites. Necropsy: hepatomegaly + hemorrhage.	Classic history = breeding facility or exotic pet show exposure; birds not yet fully feathered. Basophilic intranuclear inclusion bodies are the hallmark.
Pacheco's Disease	Herpesvirus	Sudden death in young birds with hepatic necrosis . Often no premonitory signs.	Think herpes any time you see peracute death + liver necrosis in psittacines.
Proventricular Dilatation Disease ("macaw wasting disease")	Avian bornavirus	Weakness, ataxia (it is a ganglioneuritis — GI and CNS), weight loss despite eating, regurgitation , undigested seed in droppings.	Supportive care only. Radiographs show a dilated proventriculus.

POLYOMAVIRUS VS. PBFD — THE DISCRIMINATOR

Sudden death in nestling budgies/psittacines from a breeding facility, hepatomegaly + hemorrhage on necropsy → Polyomavirus.

Progressive, symmetric feather loss and abnormal regrowth over months in a young psittacine, eventually with recurrent infections → PBFD (circovirus).

Mnemonic: Polyoma = Peracute. Circovirus = Chronic.

1.2 Bacterial Disease

Disease	Agent	Signs	Diagnosis & Treatment
Psittacosis / "Parrot fever" / Ornithosis / Avian chlamydiosis R Z	Chlamydia psittaci — obligate intracellular, gram-negative (stains pink); forms intracytoplasmic inclusions	Non-specific URI signs; biliverdinuria (lime-green urates — the buzzword); conjunctivitis; sinusitis; hepatosplenomegaly ; green-yellow diarrhea; weight loss; ruffled feathers; death. Affects all birds .	Doxycycline (drug of choice, 45-day course) ± tobramycin . Reportable and zoonotic — causes atypical pneumonia in humans, especially bird owners and pet-store workers.
Pododermatitis ("bumblefoot")	Staphylococcus (S. aureus)	Swollen, hot, painful footpad with a caseous core and scab . Predisposed by hard flooring, wet substrate, obesity, and small perch diameter in raptors .	Mild: topical antibiotics + bandaging. Moderate/severe: surgical debridement + systemic enrofloxacin. Osteomyelitis = poor prognosis .

Disease	Agent	Signs	Diagnosis & Treatment
Avian mycobacteriosis / TB Z	Mycobacterium avium subsp. <i>avium</i>	Gradual weight loss, weakness, granulomas ; lesions on spleen and liver . Classic cause of TB in cranes .	Zoonotic, especially to immunocompromised people. Treatment generally not attempted in flocks.
Infectious sinusitis	Mycoplasma gallisepticum	Chronic sinusitis, swollen infraorbital sinuses, ocular/nasal discharge.	PCR/serology; tylosin or tetracyclines.

1.3 Fungal Disease

Disease	Agent	Signs	Treatment
Aspergillosis	<i>Aspergillus fumigatus</i>	Two presentations to remember: (1) pneumonia / air sacculitis and (2) encephalopathy . Voice change, dyspnea, tail bob. Predisposed by stress, poor ventilation, moldy substrate.	Itraconazole (avoid in African greys — sensitive). Nebulized terbinafine/F10 adjunctive.
Candidiasis	<i>Candida albicans</i>	Thick white raised pseudomembrane / plaques of the crop and esophagus . Common in hand-fed babies and after antibiotic use.	Nystatin (oral, non-absorbed). Correct husbandry/antibiotic overuse.
Histoplasmosis	<i>Histoplasma capsulatum</i>	Lesions on the ceca and liver . Birds (and their droppings) act as environmental sources for dogs, cats, and humans.	Environmental/public health significance > clinical avian disease.
Cryptococcosis Z	<i>Cryptococcus neoformans</i>	Strongly associated with pigeon droppings . Birds serve as environmental hosts/carriers.	Key public-health link — CNS cryptococcosis in immunocompromised humans and in cats.

1.4 Parasitic Disease

Parasite	Presentation	Diagnosis & Treatment
Trichomonas gallinae ("canker" / "frounce" in raptors)	White-to-yellow plaques in the mouth, crop, esophagus , and extending to liver. Classic in pigeons/doves and the raptors that eat them.	Wet mount of crop washing → motile flagellates. Metronidazole. Control = hygiene and disinfection.
Coccidia — <i>Isospora</i> and <i>Eimeria</i>	Malabsorption and mucosal damage → diarrhea . Affects young, stressed, overcrowded birds.	Fecal float. Sulfonamides .
Knemidocoptes — K. pilae (scaly face/beak, budgies & passerines), K. mutans (scaly leg)	Honeycombed, proliferative crusts on the cere, beak, face, and distal extremities/legs . Seen in young or old birds with immunosuppression .	Skin scrape. Ivermectin .
Capillaria contorta	Infects the crop (and oral cavity/esophagus).	Fecal; benzimidazoles.
Syngamus trachea ("gapeworm")	Infects the trachea → gasping/gaping, head shaking.	Tracheal Y-shaped red worms; benzimidazoles.
Leucocytozoon	An RBC parasite of birds causing anemia (see full entry in the poultry chapter — infects RBCs <i>and</i> WBCs).	Blood smear; sulfonamides + pyrimethamine.

1.5 Nutritional & Metabolic Disease

Condition	Pathogenesis & Signs	Key Points
Hypovitaminosis A	Squamous metaplasia of epithelial tissue → sinusitis, chronic "big eye," white plaques around the face and in the oral cavity , blunted choanal papillae.	Classic all-seed-diet psittacine. The mimic for candidiasis/trichomoniasis plaques — always ask about diet .
Goiter (iodine deficiency / thyroid hyperplasia)	Affects older birds ; the joints involved are the metatarsal and phalangeal joints . Classic budgie presentation is regurgitation and a squeaky respiratory click from thyroid compression of the syrinx.	All-seed diets (millet/sunflower) are iodine deficient. Supplement iodine.
Xanthoma	Fatty, yellow mass in the subcutis — a lipid-laden granulomatous reaction, not a true neoplasm.	Classic in cockatiels and parakeets , often over the wing/sternum. Surgical excision.

1.6 Husbandry & Trauma

Condition	Key Facts
Crop burn	Thermal damage from hand-rearing formula fed too hot . Causes feather loss and dark, scabby, wet, oozy skin extending from the neck to the thorax ; eventually a crop-to-skin fistula. Delay surgical repair until the burn has fully demarcated.
Ingluviotomy	Surgical opening of the crop of a bird to remove a foreign body (or place a feeding tube).

RAPID-FIRE PET BIRD RECALL

Cauliflower plaques in the mouth & crop → **Candida** (nystatin) or **Trichomonas** or **hypovitaminosis A** — those are your three.

Pneumonia *and* encephalitis in a bird → **Aspergillus**.

Fatty SQ mass in a cockatiel → **xanthoma**.

Sudden death, young bird, liver necrosis → **Pacheco's (herpes)**.

Lime-green urates → **Chlamydia psittaci**.

Regurgitating macaw wasting away → **PDD (bornavirus)**.

2 | Poultry, Turkeys & Gamebirds

THE SINGLE HIGHEST-YIELD POULTRY DISCRIMINATION ON THE NAVLE

Marek's vs. Lymphoid Leukosis vs. Riboflavin deficiency — all three give you a lame chicken with nerve or tumor findings. Age, symmetry, and bursal involvement separate them. Full table in §2.6.

2.1 Viral Disease

Disease	Agent	Signs / Lesions	Diagnosis & Control
Marek's Disease	Herpesvirus (alphaherpesvirus)	Chicken <16 weeks. Asymmetric leg paralysis — " one leg forward, one leg back " (pathognomonic posture). Gray eye / gray iris (pathognomonic). Enlarged sciatic nerve with lymphocytic infiltration. Lymphoid tumors in multiple organs . Also associated with atherosclerosis.	Horizontal transmission via feather dander . Vaccinate at hatch . No treatment .
Lymphoid Leukosis	Retrovirus — Avian Leukosis Virus (ALV)	Chicken >16 weeks. Bursal lymphoma ; enlarged liver and spleen. No nerve enlargement , no early mortality.	Bursa of Fabricius = target organ . Transmitted vertically through the egg . Histopath shows uniform large lymphoblasts. No treatment .
Infectious Bursal Disease (Gumboro) R	Birnavirus	Enlarged bursa ; immunosuppression via destruction of B cells ; bursa of Fabricius, thymus, and spleen affected.	Shed in feces , spread by fomites — highly contagious and environmentally hardy. Vaccinate; depopulate heavily contaminated houses. The reportable immune-system disease of birds .
Avian Encephalomyelitis (Epidemic Tremor)	Picornavirus	Chicks <3 weeks: progressive ataxia, head and neck tremors , dull sunken eyes, sitting on hocks . Blue opacity of the lens . Adult hens: transient egg drop .	Egg-transmitted (vertical) — the classic feature . Histopath: lymphocytic infiltration of the proventriculus + CNS neuronal degeneration. No treatment; vaccinate pullets 4–8 weeks before lay for maternal antibody.
Avian Viral Arthritis / Tenosynovitis	Reovirus	Broiler with swollen hock joints , lameness, sitting on hocks, gastrocnemius tendon swelling ; chronic cases → tendon rupture .	Virus isolation or PCR from joint/tendon . No treatment; vaccinate breeder hens for maternal antibody. Differentiate from Mycoplasma synoviae by M. synoviae's silver/gray exudate + respiratory signs .
Fowl Pox (Avian Pox)	Avipoxvirus	Cutaneous / "dry" form : scabby, proliferative, ulcerative lesions on comb, wattles, and legs — low mortality . Diphtheritic / "wet" form : yellow canker lesions on mucous membranes — tongue, esophagus, trachea — higher mortality (airway obstruction).	Mosquito-transmitted and by direct contact through skin abrasions. Vaccination (wing-web stab).
Infectious Bronchitis	Coronavirus	Wrinkled, misshapen eggs and decreased egg production/quality; respiratory signs; nephropathogenic strains. High morbidity, low mortality .	Antibiotics only for secondary bacterial infection. Vaccination.
Infectious Laryngotracheitis (ILT) R	Herpesvirus	Blood-stained beaks and gasping for air — the buzzword pair. Bloody mucus in the trachea.	Reportable . Intranuclear inclusions in tracheal epithelium.

Disease	Agent	Signs / Lesions	Diagnosis & Control
Blue Comb / Coronaviral enteritis of turkeys	Coronavirus	Affects turkeys . Diarrhea and dehydration; cyanotic ("blue") comb/head from dehydration and shock.	Supportive care, warmth, electrolytes.
Marble Spleen Disease / Hemorrhagic Enteritis	Fowl Adenovirus Group II (siadenovirus)	Young pheasant found dead with a massively enlarged, mottled, marble-like spleen → Marble Spleen Disease. Same virus causes Hemorrhagic Enteritis in turkeys .	Gross pathology + intranuclear inclusion bodies on histopath. Live vaccine available .
Swollen Head Syndrome / Turkey Rhinotracheitis	Avian metapneumovirus (enveloped paramyxovirus)	Swollen head in chickens ; rhinotracheitis in turkeys . Periorbital and facial edema.	One virus, two species names — a favorite trick question.
Newcastle Disease (velogenic / virulent — vND) 	Avian paramyxovirus-1	Petechial hemorrhages on the mucosa of the proventriculus and gizzard (the classic lesion) and petechiae on GI serosa. Respiratory, neurologic, and GI signs together.	Reportable — federal . Diagnose by virus isolation/RT-PCR.
Highly Pathogenic Avian Influenza (HPAI)  	Influenza A	Cyanosis , edema (head/wattles), SQ ecchymoses , hemorrhages, torticollis , opisthotonos , green diarrhea , sudden high mortality.	RT-PCR of choanal cleft and tracheal swabs . Immediately reportable; depopulation.

2.2 Bacterial Disease

Disease	Agent	Signs / Lesions	Treatment & Control
Fowl Typhoid	Salmonella gallinarum	High mortality in growing/adult birds; bronze-green liver.	Both S. gallinarum and S. pullorum cause high mortality in chickens and are the two classic host-adapted, egg-transmitted salmonellae. Test-and-slaughter eradication programs (NPIP).
Pullorum Disease	Salmonella pullorum	Five signs to know: high mortality , unabsorbed yolk sac , anorexia, cecal cores , diarrhea. Disease of young chicks.	
Colibacillosis	E. coli	Subacute pericarditis , air sacculitis , salpingitis , and fatal septicemia. (Contrast: enterotoxemia is clostridial.)	Often secondary to viral/ mycoplasmal respiratory disease or poor air quality.
Salpingitis	Ascending E. coli	Inspissated (cheesy) material within the oviduct . Predisposed by excessive body condition and poor coop hygiene .	Prevention through BCS control and litter management.
Fowl Cholera	Pasteurella multocida	Fibrinous polyserositis , acute septicemia, swollen wattles , torticollis (chronic localized form).	Tetracycline . Rodent and wildlife reservoirs.
Infectious Coryza	Avibacterium paragallinarum (formerly <i>Haemophilus</i>)	Swelling under the eye (infraorbital sinus), sneezing, nasal discharge, foul odor.	All-in / all-out management is the control answer. Sulfonamides/ tetracyclines reduce signs but carriers persist.
Bordetellosis / Turkey Coryza	Bordetella avium — gram-negative coccobacillus	Turkey poults 1–6 weeks : watery eyes, nasal discharge, sneezing, foamy eye, and tracheal collapse with a clicking breathing sound (the buzzword). Destroys tracheal cilia.	Culture from tracheal swab . Tetracycline/sulfonamides for secondary infections. Vaccinate poults at 1–2 weeks with live attenuated vaccine in the drinking water .

Disease	Agent	Signs / Lesions	Treatment & Control
Ulcerative Enteritis ("quail disease")	<i>Clostridium colinum</i>	Humped back, ruffled feathers, bloody-to-watery white droppings with pseudomembrane , ulcerative enteritis.	Compare with <i>C. perfringens</i> necrotic enteritis → "Turkish towel" gut .
Necrotic Enteritis	<i>Clostridium perfringens</i>	"Shaggy gut" / Turkish-towel mucosa. Predisposed by coccidiosis damaging the mucosa first.	Control coccidia to control necrotic enteritis. Bacitracin/penicillins in feed (where permitted).
Bacterial synovitis	<i>Staphylococcus</i> (most common infectious agent)	Swollen joints, lameness.	Also consider <i>Mycoplasma synoviae</i> and reovirus.
Avian Tuberculosis Z	<i>Mycobacterium avium</i> subsp. <i>avium</i>	Gradual weight loss and granulomas , especially liver, spleen, and intestine.	Zoonotic . Depopulation of infected flocks.
Erysipelas Z	<i>Erysipelothrix rhusiopathiae</i>	Sudden death, especially in turkeys; septicemia.	Penicillin . (Note the contrast with laying/broiler restrictions below.)
Avian Chlamydiosis R Z	<i>Chlamydia psittaci</i>	In poultry also called parrot fever . Conjunctivitis, sinusitis, green-yellow diarrhea, weight loss, ruffled feathers, hepatosplenomegaly . Intracellular gram-negative (pink) with intracytoplasmic inclusions .	Doxycycline ± tobramycin . Zoonotic, reportable, affects all birds .
Fowl spirochetosis	<i>Borrelia anserina</i>	Septicemia, green diarrhea, splenomegaly.	Vector = Argas persicus (fowl tick — the soft tick).

2.3 Parasitic Disease

Protozoa

Parasite	Presentation	Diagnosis & Treatment
Coccidiosis — <i>Eimeria</i> spp.	Four classic signs: cheesy cecal/cloacal cores, slicked-over litter, severe diarrhea , and necrotic enteritis / "shaggy gut." Necropsy: necrotic, hemorrhagic intestines . High mortality; the most important parasitic disease of chickens.	Amprolium (a thiamine analog) is the treatment; reduce hen access to feces is the control answer. Ionophores/anticoccidials in feed.
Eimeria tenella	The species found in the ceca → cecal cores, blood in the cecum , and hemorrhagic droppings .	Memorize: tenella = ceca = blood .
Histomoniasis ("Blackhead")	Histomonas meleagridis . Signs: bull's-eye ("target") lesions on the liver , listlessness, cecal changes/cores, sulfur-yellow droppings , unkempt feathers. Turkeys most severely affected.	Carried by Heterakis gallinarum (cecal worm) — and the <i>Heterakis</i> eggs are in turn carried by earthworms. Control the cecal worm; don't co-house chickens (carriers) with turkeys.
Leucocytozoonosis	Young ducks/turkeys/chickens near fast-moving water in spring with anemia, green diarrhea, splenomegaly, sudden death . Three sign categories: anemia, neurologic, respiratory . Infects both RBCs and WBCs (lymphocytes) .	Vectors: black flies (<i>Simulium</i>) for ducks/turkeys; <i>Culicoides</i> for chickens . Blood smear (gametocytes in RBCs/WBCs) + megaloschizonts on histopath. Treat with sulfonamides + pyrimethamine .
Avian Malaria — <i>Plasmodium</i> spp.	Penguin or raptor in captivity with anemia, weakness, splenomegaly, sudden death. Most pathogenic in non-adapted species (penguins, raptors); mild/subclinical in chickens.	Vector = mosquito . Blood smear: ring forms, schizonts, gametocytes INSIDE RBCs only . Treat with chloroquine + primaquine .
Trichomoniasis	<i>Trichomonas gallinae</i> → caseous exudate in the mouth / oral and crop lesions.	The differential for candidiasis causing caseous oral/esophageal exudate in a chicken. Control by hygiene and disinfection .

LEUCOCYTOZOOM VS. PLASMODIUM — THE ONE-LINE DISCRIMINATOR

Leucocytozoon infects **RBCs AND WBCs**; vector = **black fly** / *Culicoides*; look for **megaloschizonts**.

Plasmodium (avian malaria) infects **RBCs ONLY**; vector = **mosquito**; classic in **penguins and raptors**.

Helminths

Worm	Location / Effect	Treatment
The three roundworms of poultry: <i>Ascaridia</i> , <i>Heterakis</i> , <i>Capillaria</i>		
Ascaridia galli	Small intestine — the large roundworm; unthriftiness, obstruction in heavy burdens.	Fenbendazole treats <i>Ascaridia</i> , <i>Heterakis</i> , and <i>Capillaria</i> .
Heterakis gallinarum	Ceca — largely non-pathogenic itself, but carries <i>Histomonas</i> (blackhead). That's why it matters.	
Capillaria	Depending on species: crop/esophagus (<i>C. contorta</i>) or intestine; also lung, bladder, liver forms in other hosts.	
Syngamus trachea (gapeworm)	Trachea → gaping, gasping, head shaking.	Benzimidazoles; earthworm paratenic hosts.
Davainea proglottina	Tapeworm of the duodenum of birds . Slug/snail intermediate hosts.	Praziquantel; control molluscs.

Ectoparasites

Parasite	Key Feature	Treatment
Northern fowl mite <i>Ornithonyssus sylviarum</i>	Lives on the bird ; found during the DAY , near the vent area . Blackened, matted vent feathers.	Permethrin dusts/sprays applied to birds.
Red mite / roost mite <i>Dermanyssus gallinae</i>	Lives off the bird in cracks and crevices; feeds at NIGHT ; sucks blood → anemia, drop in production.	Miticide applied to the environment/housing , not just the birds.
Scaly leg mite <i>Knemidocoptes mutans</i> / <i>K. pilae</i>	Burrows into the skin of the distal extremities → raised, crusted, honeycombed scales. <i>K. pilae</i> = scaly face of budgies/passerines.	Ivermectin .
Argas persicus (fowl tick)	Soft tick; vector for fowl spirochetosis .	Environmental acaricide.

NORTHERN FOWL MITE VS. RED MITE

Northern fowl mite = **on** the bird, seen by **day**, at the vent → treat the **bird**.

Red mite (*Dermanyssus*) = in the housing, feeds at **night** → treat the **house**.

Fungal

Disease	Presentation	Diagnosis & Treatment
Favus ("White Comb")	Essentially ringworm of poultry , caused by <i>Trichophyton (Lophophyton) gallinae</i> . White, powdery, crusting lesions on the comb and wattles. NOT pruritic .	Definitive diagnosis by culture (KOH prep or DTM). Treat with miconazole .
Candidiasis	Mycotic disease with a thick, white, raised pseudomembrane of the crop and esophagus .	Copper sulfate in drinking water is the flock-level answer; nystatin for individuals.
Aspergillosis ("brooder pneumonia")	Air sacculitis and pneumonia. Infected eggs can explode in the incubator (a classic hatchery question).	Hatchery sanitation; discard moldy litter/feed.

2.4 Nutritional, Metabolic & Production Disease

Condition	Key Facts
Riboflavin (Vitamin B₂) deficiency	1–2 week old chick with bilateral CURLED TOES , symmetric leg weakness, enlarged, demyelinated sciatic (and brachial) nerves , NO eye lesions , NO tumors . Looks like Marek's but is symmetric and occurs at 1–2 weeks of age . Fully reversible with immediate riboflavin supplementation if caught early.
Ascites Syndrome (pulmonary hypertension syndrome)	Fast-growing broilers outgrow their cardiopulmonary capacity. Confirmatory lesion = right ventricular hypertrophy . Abdominal fluid, cyanosis. Manage by slowing growth rate, improving ventilation/oxygenation, reducing altitude stress.
Dissecting aneurysm of turkeys	Rapidly grown turkey with acute internal hemorrhage and sudden death — rupture of the aorta. Associated with rapid growth and copper deficiency/lathyrogens.
Round Heart Disease	Affects turkeys ; a spontaneous cardiomyopathy with a globoid, dilated heart.
Visceral Gout	Urate deposits on the pericardium, liver, and kidney . Driven by renal failure — high dietary calcium, dehydration, nephropathogenic IBV, or excess protein.
White Muscle Disease	Vitamin E and selenium deficiency — affects poultry as well as pigs, small ruminants, and horses. Pale, chalky skeletal and cardiac muscle.

2.5 Poultry Pharmacology & Residues

EXTRALABEL DRUG RESTRICTIONS IN POULTRY

The federal rule (21 CFR 530.41) prohibits **extralabel use of fluoroquinolones (enrofloxacin) and glycopeptides in poultry**, and prohibits **extralabel use of cephalosporins** in major food species. Extralabel drug use in **laying hens producing eggs for human consumption** is effectively off-limits because no withdrawal data exist. Learn it as: **no ELDU in layers; no fluoroquinolones in poultry**.

Drug	Poultry Use
Amprolium	Coccidiosis (thiamine analog — competitive inhibitor of thiamine uptake by the coccidia).
Fenbendazole	<i>Ascaridia</i> , <i>Heterakis</i> , <i>Capillaria</i> .
Copper sulfate	Candidiasis in drinking water.
Tetracycline	Fowl cholera; chlamydiosis (doxycycline preferred); <i>Bordetella avium</i> secondaries.
Penicillin	Erysipelas. Restricted/prohibited from extralabel use in laying and broiler birds.
Sulfonamides + pyrimethamine	Leucocytozoonosis.
Chloroquine + primaquine	Avian malaria.
Miconazole	Favus.

2.6 The Lame Chicken Algorithm

Feature	Marek's Disease	Lymphoid Leukosis	Riboflavin Deficiency	Avian Viral Arthritis
Agent	Herpesvirus	Retrovirus (ALV)	Nutritional (B ₂)	Reovirus
Age	<16 weeks	>16 weeks	1–2 weeks	Broilers, growing
Symmetry	Asymmetric — one leg forward, one back	n/a (tumor disease)	Symmetric , curled toes	Bilateral hocks
Nerves	Enlarged , lymphocytic infiltration	NOT enlarged	Enlarged, demyelinated	Normal
Eye	Gray iris (pathognomonic)	Normal	No eye lesions	Normal

Feature	Marek's Disease	Lymphoid Leukosis	Riboflavin Deficiency	Avian Viral Arthritis
Tumors	Multiple organs	Bursa of Fabricius is the target	None	None
Transmission	Horizontal — dander	Vertical — through the egg	Diet	Horizontal + vertical
Control	Vaccinate at hatch ; no treatment	No treatment; eradication programs	Supplement — reversible	Vaccinate breeder hens

RAPID-FIRE POULTRY RECALL

Marble spleen in a pheasant → **Fowl adenovirus II** (= hemorrhagic enteritis of turkeys).

Clicking breath + tracheal collapse in a poult → **Bordetella avium**.

Wrinkled eggs → **Infectious bronchitis (coronavirus)**.

Bloody beak + gasping → **ILT**.

Blue comb in turkeys → **coronavirus**.

Bull's-eye liver lesion + yellow droppings → **Blackhead** (*Histomonas*, carried by *Heterakis*).

Swollen head in chickens / rhinotracheitis in turkeys → **avian metapneumovirus**.

Swelling under the eye + sneezing → **Infectious coryza** (*Avibacterium paragallinarum*) → **all-in/all-out**.

Right ventricular hypertrophy in a broiler → **ascites syndrome**.

Shaggy gut → **coccidiosis predisposing to *C. perfringens***.

3 | Avian Cross-Cutting Concepts

3.1 Anatomy & Physiology You Will Be Asked About

Concept	What to Know
Avian erythrocytes	Nucleated, elliptical (oval) RBCs. Same for reptiles, amphibians, and fish. Thrombocytes replace platelets, and heterophils replace neutrophils.
Pneumatized bones	The humerus and femur are pneumatized — they communicate with the air sac system. Never place an intraosseous catheter in the humerus or femur (you will drown the bird by infusing fluid into the air sacs). Use the ulna or the tibiotarsus instead.
Two-part stomach	Proventriculus = glandular stomach (the true stomach; dilated in PDD, hemorrhagic in Newcastle). Ventriculus (gizzard) = muscular grinding stomach.
Crop (ingluvies)	Esophageal storage diverticulum. Ingluviotomy = surgical opening of the crop. Crop stasis is a red flag for polyomavirus, PDD, candidiasis, and foreign bodies.
Bursa of Fabricius	Site of B-cell maturation. Target of IBD/Gumboro (destruction → immunosuppression) and lymphoid leukemia (neoplastic transformation).

3.2 Any Bird, Any Species — Cross-Species Differentials

Clinical Picture	Differential List
White/yellow plaques in the mouth, crop, or esophagus	1. Candidiasis (nystatin / copper sulfate) · 2. Trichomoniasis (<i>T. gallinae</i> , metronidazole) · 3. Hypovitaminosis A (squamous metaplasia) · 4. Wet/diphtheritic fowl pox (yellow canker lesions).
Pneumonia and/or encephalitis in a bird	Aspergillus — this pair of organ systems in one bird is the giveaway.
Sudden death in young birds	Pacheco's disease (herpes, liver necrosis) · Polyomavirus (SQ hemorrhage, crop stasis) · septicemia.
Nestling mortality with SQ hemorrhages	Polyomavirus — young bird with crop stasis, not fully feathered, exotic pet show or breeder exposure.
Chronic sinusitis, "big eye," white plaques around the face	Vitamin A deficiency.
Anemia from an RBC parasite of birds	Leucocytozoon (RBCs + WBCs, black fly / <i>Culicoides</i>) vs. Plasmodium (RBCs only, mosquito, penguins/raptors).
Swollen, hot, painful footpad with a caseous core	Bumblefoot — <i>Staphylococcus aureus</i> ; chicken, turkey, or raptor.
Scaly, crusted face or legs	Knemidocoptes — <i>K. pilae</i> (face/beak, budgies & passerines), <i>K. mutans</i> (legs, poultry). Ivermectin.
Zoonotic + reportable respiratory disease of any bird	Chlamydia psittaci — psittacosis/ornithosis/parrot fever. Doxycycline.

3.3 Toxicology Relevant to Birds

Toxin	Effect
Blue-green algae (cyanobacteria)	Birds and cattle are the most commonly affected species. Microcystin → acute hepatotoxicity; anatoxin-a → neurotoxicity and rapid death. Stagnant, warm, nutrient-rich water in late summer.
Heavy metals (lead, zinc)	Classic caged-bird toxicoses from galvanized wire, curtain weights, costume jewelry, and pennies. Signs: regurgitation, neurologic disease, hemoglobinuria (green droppings). Chelate with CaEDTA.

AVIAN REPORTABLE/ZOONOTIC LIST — MEMORIZE AS A BLOCK

Reportable: HPAI · Virulent Newcastle Disease (vND) · Infectious Laryngotracheitis · Infectious Bursal Disease · Avian Chlamydiosis.

Zoonotic: Chlamydia psittaci · Avian influenza · *Mycobacterium avium* · *Cryptococcus* (pigeon droppings) · *Salmonella* · *Campylobacter* · Histoplasma (environmental).

4 | Fish & Aquaculture

THE FISH EXAM MINDSET

For any sick-fish question, work the list in this order: **(1) water quality** → **(2) parasites** → **(3) bacteria** → **(4) virus**. The NAVLE overwhelmingly tests water quality and the handful of reportable/foreign animal aquatic diseases.

4.1 Water Quality — The First Differential Always

Parameter / Problem	Pathophysiology & Signs	Management
Ammonia — ionized vs. un-ionized	Ionized ammonia = NH_4^+ = relatively harmless . Un-ionized ammonia = NH_3 = highly toxic → cellular damage and suffocation. The NH_3 fraction rises as pH and temperature rise.	Water changes, reduce feeding/stocking, buffer pH downward carefully, establish biofiltration.
pH	Ideal range for most freshwater fish = 6.5–8.5 . As pH increases (more alkaline), fish become hyperactive and ammonia shifts toward the toxic un-ionized NH_3 form.	Buffer with carbonate hardness; avoid rapid swings.
"New tank syndrome"	NITRITE accumulation because the biofilter is not established. Nitrite oxidizes hemoglobin → methemoglobin → dark brown gills ("brown blood disease") . Separately, nitrate toxicity causes immune suppression .	Cycle the tank before stocking; add salt (chloride competes with nitrite uptake at the gill); water changes.
Gas supersaturation	Water contains more dissolved gas than it can naturally hold at a given temperature and pressure → bubbles form in the blood and tissues = Gas Bubble Disease . Exophthalmos, bubbles in fins and gill lamellae.	Find the leak in the plumbing on the suction side; degas with packed columns/aeration.
Hyposalinity	Associated with swim bladder trauma, metabolic acidosis, and internal hemorrhage .	Correct salinity gradually.
Chlorine / chloramine	Erratic swimming and excessive mucus on the gills . Municipal tap water added without dechlorination.	Sodium thiosulfate; always dechlorinate.
Copper	Copper is more toxic in soft water and acidic water (less binding, more free cupric ion). Narrow therapeutic index when used as a parasiticide.	Measure free copper; monitor alkalinity/hardness.
Zinc	Zinc inhibits calcium uptake at the gill. Sources include galvanized fittings.	Avoid galvanized plumbing.
"Penny toxicity"	Caused by pennies thrown into a pond or tank — a zinc (and copper) toxicosis.	Remove the coins; the classic koi-pond history.
Water too cold	Predisposes to <i>Saprolegnia</i> (a water mold) infection.	Identify on direct wet mount ; treat with formalin or hydrogen peroxide . Correct the temperature.

4.2 Viral Disease

Disease	Agent	Signs / Species	Diagnosis & Action
Tilapia Lake Virus (TiLV) 	Enveloped orthomyxovirus (emerging)	10–90% mortality . Pale gills, dark ulcerated skin, fluid in the coelom .	WOAH-listed / reportable . Call the authorities and depopulate.
Infectious Salmon Anemia (ISA) 	Orthomyxovirus (isavirus)	Severe anemia; kidney and liver necrosis with abdominal edema ; pale gills; lethargy. Farmed Atlantic salmon.	Reportable — no treatment . Eradicate the entire stock .

Disease	Agent	Signs / Species	Diagnosis & Action
Infectious Hematopoietic Necrosis (IHN) 	Novirhabdovirus (rhabdovirus)	Salmonids. Severe anemia, fluid accumulation, hemorrhage, pale gills , exophthalmia , distended abdomen . Most susceptible at ~2 months / fry-fingerling stage.	Endemic in the Pacific Northwest and Alaska. Reportable; no treatment.
Megalocytivirus — Infectious Spleen & Kidney Necrosis Virus / Scale Drop Disease 	Iridoviridae	Splenic and renal necrosis; scale loss; high mortality in cultured marine and ornamental fish.	Diagnose by PCR of SPLEEN and KIDNEY. Report to authorities.
Lymphocystis	Iridoviridae	Cauliflower-like masses on the fins that "look like balloons" — hypertrophied dermal fibroblasts. Cosmetic; usually self-limiting.	Do nothing — the answer is no treatment. Reduce stress; lesions regress.

4.3 Bacterial Disease

Disease	Agent	Signs / Species	Management
Furunculosis / "goldfish ulcer disease"	<i>Aeromonas salmonicida</i> — gram negative	Trout and farmed salmon — the most common bacterial disease of farmed salmon . Necrotic hemorrhages on the tail, fins, gills, and internal organs , progressing to kidney and spleen . Boil-like furuncles in the musculature. Also the cause of goldfish ulcer disease.	Culture; florfenicol/ oxytetracycline in feed; vaccination in aquaculture.
Motile aeromonad septicemia	<i>Aeromonas hydrophila</i> and relatives	Opportunistic pathogen associated with STRESS — the exam's favorite framing. Ulcers, hemorrhage, dropsy.	Fix the underlying stressor (water quality, crowding, handling).
Enteric Septicemia of Catfish (ESC)	<i>Edwardsiella ictaluri</i> — gram negative	Hemorrhagic enteritis and "hole in the head" lesion (open lesion over the frontal fontanelle). The major disease of channel catfish aquaculture.	Medicated feed (florfenicol); vaccination; manage water temperature.
Columnaris	<i>Flavobacterium columnare</i>	"Saddleback" lesions, frayed fins, gill necrosis, cotton-like mouth lesions. Classically a WARM-water disease — outbreaks are worst in summer, above roughly 15–20°C; virulence is strain- and temperature-dependent. (The cold-water look-alike is <i>Saprolegnia</i> .)	Prevention: vaccination is available; reduce organic debris and prevent trauma. Salt and potassium permanganate baths.

4.4 Parasitic Disease

Parasite	Presentation	Diagnosis & Treatment
Sea lice — <i>Lepeophtheirus salmonis</i>	Copepod ectoparasite of farmed salmon ; grazing erosion of skin, secondary infection, and huge economic loss.	Emamectin benzoate, hydrogen peroxide baths, cleaner fish, mechanical delousing.
Marine white spot disease = SALTWATER ICH — <i>Cryptocaryon irritans</i>	White spots on skin and gills; flashing, respiratory distress.	ID trophonts — ciliated, pear-shaped parasites — on a wet mount from a skin scrape . Copper or hyposalinity therapy.
Amyloodinium = "Velvet"/ coral fish disease	Parasitic dinoflagellate . Skin, fins, and gills appear velvety (gold dust). Rapid gill mortality in marine systems.	Copper sulfate, hydrogen peroxide, or chloroquine.

Parasite	Presentation	Diagnosis & Treatment
Whirling disease — <i>Myxobolus cerebralis</i>	Myxozoan parasite of salmonid cartilage → skeletal deformity, blacktail, and the characteristic whirling swimming behavior. Tubifex worm intermediate host.	Identify by PCR. No effective treatment; control by avoiding earthen ponds and disinfecting.
Bolbophorus	Digenean trematode of catfish in the southern/southwestern US. Causes red and white raised bumps in the muscle and fluid in the abdomen . Mortality up to 95%.	Life cycle involves the American white pelican and rams-horn snails — control by bird deterrence and snail control.
Microsporidiosis — <i>Glugea</i>	Xenoma formation; chronic wasting and visible white nodules.	Depopulate and disinfect — there is no practical treatment.

4.5 Fungal & Oomycete Disease

Disease	Presentation	Treatment
<i>Saprolegnia</i> (water mold)	Cotton-wool growth on skin/gills. Follows cold water , trauma, or immunosuppression.	Formalin or hydrogen peroxide. Identify on direct wet mount . Correct temperature and husbandry.
Epizootic Ulcerative Syndrome (EUS) 	Ulcerative skin disease — ulcers with RED CENTERS and WHITE RIMS. Necropsy: soft tissue granulomas and liquefactive necrosis of muscle.	Contact the federal authorities — this is a foreign animal / WOAHA-listed disease. Caused by the oomycete <i>Aphanomyces invadans</i> .
<i>Fusarium solani</i>	Fungal disease of captive marine fish (also elasmobranchs and crustaceans). Skin ulcers and systemic granulomas.	Poor prognosis; antifungals of limited value.

THE TWO CAUSES OF ANEMIA IN FISH

- Infectious Salmon Anemia virus** — an orthomyxovirus; reportable, no treatment.
- Nitrite toxicity ("brown blood disease")** — or blood-feeding ectoparasites such as sea lice and *Argulus*.

4.6 Nutritional Disease

Condition	Key Facts
"Broken back disease"	Ascorbic acid (vitamin C) deficiency → defective collagen, scoliosis/lordosis, spinal deformity, impaired wound healing. Vitamin C is heat-labile and degrades in stored feed.

4.7 Neoplasia

Tumor	Species
Fibrosarcoma	The most common neoplasia of goldfish . Also consider lymphocystis as the viral non-neoplastic mimic of a "mass" on a fin.

4.8 Anesthesia, Euthanasia & Residues

Agent	Use & Key Numbers
MS-222 (tricaine methanesulfonate)	The standard fish anesthetic and euthanasia agent. Withdrawal time = 21 days. Must be buffered with sodium bicarbonate (it is acidic in solution). Immersion route.
Aquacalm (metomidate hydrochloride)	Anesthetic/sedative used in ornamental fish — particularly useful for transport sedation. Not for food fish.

AQUATIC REPORTABLE DISEASES — KNOW THIS LIST COLD

Tilapia Lake Virus · Infectious Salmon Anemia · Infectious Hematopoietic Necrosis · Megalocytivirus (ISKNV / scale drop disease) · Epizootic Ulcerative Syndrome.

For every one of these, the correct exam answer includes **notify state/federal authorities**, and usually **depopulate — there is no treatment**.

RAPID-FIRE FISH RECALL

Cauliflower/balloon masses on fins → **lymphocystis (iridovirus)** — **do nothing**.

Dark brown gills in a new tank → **nitrite**.

Erratic swimming + excess gill mucus → **chlorine/chloramine**.

Velvety gold skin → **Amyloodinium** → copper/H₂O₂/chloroquine.

Pear-shaped ciliated trophonts on skin scrape in a marine tank → **saltwater ich (Cryptocaryon)**.

Red-centered, white-rimmed ulcers → **EUS — call the feds**.

Hole in the head + hemorrhagic enteritis in catfish → **Edwardsiella ictaluri**.

Red/white bumps in catfish muscle + ascites → **Bolbophorus**.

Cotton wool after the heater failed → **Saprolegnia** → formalin or H₂O₂.

5 | Reptiles & Amphibians

THE REPTILE MINDSET

Reptile medicine on the NAVLE is **husbandry medicine**. Temperature gradient, humidity, UVB, and diet explain most of what you will be shown. Only a handful of specific agents get tested.

5.1 Reptile Infectious Disease

Condition	Agent	Presentation	Treatment
Upper respiratory tract disease (URTD)	<i>Mycoplasma</i> (<i>M. agassizii</i>) — also herpesvirus and ranavirus	Turtle/tortoise with swollen eyes, nasal discharge, and lethargy. The classic chelonian presentation. Chronic, often lifelong carriage.	Supportive care, raise ambient temperature, systemic antibiotics (enrofloxacin); PCR/serology. Do not release affected wild tortoises.
Amoebiasis	<i>Entamoeba invadens</i>	Severe necrotizing colitis and hepatitis in snakes and lizards; often fatal. Turtles are frequently asymptomatic carriers — never house turtles with snakes.	Metronidazole.
Ulcerative dermatitis ("scale rot")	Gram-negative opportunists (<i>Aeromonas</i> , <i>Pseudomonas</i>)	Ulcerated, blistered ventral scales in snakes . Caused by excessive humidity, damp/dirty substrate, and poor hygiene.	Topical and systemic antibiotics PLUS correction of husbandry — the husbandry half is the exam answer.
Mycobacteriosis Z	<i>Mycobacterium</i> spp.	Associated with chronic wasting in reptiles. Granulomas in liver, spleen, lung.	Poor prognosis; zoonotic risk. Euthanasia often recommended.
Pentastomiasis Z	Pentastomes ("tongue worms" — aberrant crustaceans)	Infect reptiles and cause pneumonia. Reside in the respiratory tract.	Zoonotic — humans are accidental intermediate hosts (visceral pentastomiasis). Ivermectin and manual removal; poor response.

5.2 Reptile Ectoparasites

Parasite	Key Fact
Snake mite <i>Ophionyssus natricis</i>	The common snake mite. Transmits <i>Aeromonas hydrophila</i> — that vector role is the tested point. Mites cluster around the eyes, chin groove, and under scales; snakes soak excessively. Treat the snake and the entire enclosure (ivermectin spray on the environment — never ivermectin systemically in chelonians).

IVERMECTIN AND CHELONIANS

Ivermectin is fatal in turtles and tortoises (and in some skinks). This is a classic exam trap: an ivermectin question that is safe in a snake or lizard is lethal in a chelonian.

5.3 Amphibians

Condition	Agents	Notes
Red Leg Syndrome	Four bacteria to name: 1. <i>Aeromonas</i> (classically <i>A. hydrophila</i>) 2. <i>Pseudomonas</i> 3. <i>Proteus</i> 4. <i>Citrobacter</i>	Red spots/erythema on the ventral legs and abdomen from cutaneous hemorrhage and septicemia. Almost always secondary to poor water quality, overcrowding, or chilling . Treat the water first, then antibiotics.

RAPID-FIRE HERP RECALL

Tortoise with swollen eyes + nasal discharge → *Mycoplasma* URTD.

Snake with amoebic colitis → *Entamoeba invadens* → metronidazole.

Snake with ventral scale ulcers → **scale rot** — fix the humidity.

Snake mite → *Ophionyssus*, transmits *Aeromonas hydrophila*.

Chronic wasting reptile → *Mycobacterium*.

Reptile pneumonia from a "tongue worm" → pentastomes (zoonotic).

Frog with red legs → *Aeromonas/Pseudomonas/Proteus/Citrobacter*.

6 | Rabbits

6.1 Viral Disease

Disease	Agent	Presentation	Diagnosis & Control
Myxomatosis R	Myxoma virus (poxvirus / leporipoxvirus)	In the US, found in California, Oregon, and Baja ; carried by cottontail rabbits (the reservoir, which is subclinically infected). In domestic <i>Oryctolagus</i> : fever and acute death , plus periocular/genital edema and skin nodules. Mosquito- and flea-transmitted.	PCR . Vector control and insect-proof housing; no treatment. Vaccine available in Europe, not the US.
Rabbit Hemorrhagic Disease (RHD / RHDV-2) R	Calicivirus (lagovirus)	Two lesions to name: hepatic (liver) necrosis and bleeding — DIC with hemorrhage from the nose. Peracute death in previously healthy rabbits.	Foreign animal disease — reportable . Extremely environmentally stable; strict biosecurity. Vaccine available under emergency authorization.

6.2 Bacterial Disease

Disease	Agent	Presentation	Diagnosis & Treatment
"Snuffles"	<i>Pasteurella multocida</i> (pasteurellosis)	Serous-to-mucopurulent nasal discharge, sneezing, matted forepaws, conjunctivitis; can progress to abscesses, otitis media/interna, and pneumonia.	Diagnose by nasal culture (deep nasal swab). Enrofloxacin, trimethoprim-sulfa; carriers are common.
Rabbit Syphilis (Treponematoses / "vent disease")	<i>Treponema paraluis-cuniculi</i>	Red, ulcerated, scabby/crusting lesions on the perineum, genitalia, nose, and face . Venereally and vertically transmitted; NOT zoonotic .	PARENTERAL penicillin — NEVER oral . Three weekly injections of penicillin G benzathine/procaine.

6.3 Parasitic & Protozoal Disease

Parasite	Presentation	Diagnosis & Treatment
<i>Encephalitozoon cuniculi</i> Z	Intracellular pathogen (a microsporidian). Head tilt/ vestibular disease, posterior paresis, urinary incontinence, and phacoclastic uveitis with a white lenticular mass. Chronic interstitial nephritis.	Shed in the URINE . Treat with fenbendazole (28-day course). Zoonotic — beware in immunosuppressed owners . Serology supports but does not confirm.
Ear mites / otoacariasis <i>Psoroptes cuniculi</i>	Thick, crusted, tan-brown exudate filling the ear canal; intense pruritus and head shaking. Non-burrowing surface mite.	Selamectin (or ivermectin). Do not manually strip the crusts — it is exquisitely painful. Treat all in-contact rabbits and the environment.
Cuterebriasis ("warbles")	SQ swelling with a breathing pore, from the rodent/ rabbit botfly larva. Concern: larval migration to the CNS .	Careful whole-larva extraction (rupture → anaphylaxis). Control with antibiotics and NSAIDs . The larva has a fistulous stage.
Coccidia — <i>Eimeria</i>	<i>E. stiedae</i> = hepatic coccidiosis; intestinal <i>Eimeria</i> spp. → diarrhea in weanlings.	Eimeria has 4 sporocysts (vs. <i>Isospora</i> / <i>Cystoisospora</i> = 2). Eimeria infects ruminants, horses, rabbits, and poultry ; Isospora infects dogs, cats, and pigs . Dogs and cats can pass <i>Eimeria</i> oocysts spuriously after eating rabbit feces — a classic pseudoparasite trap.
Ringworm Z	<i>Trichophyton mentagrophytes</i> dominates in rabbits and small exotics (does not fluoresce under Wood's lamp).	Terbinafine . Zoonotic.

6.4 Anatomy, Reproduction & Husbandry

Concept	Key Facts
Dewlap	A heavy fold of skin on the ventral neck , found in intact females (and some large breeds). Predisposes to moist dermatitis ("wet dewlap") and interferes with grooming and drinking.
Placentation	Rabbits have a hemochorial, discoid placenta — the same type as humans and rodents, and the most invasive type. Contrast with the epitheliochorial, diffuse placenta of the horse and pig, where the maternal and fetal circulations remain completely separate.
Induced ovulation	The four induced ovulators to memorize: rabbits, cats, ferrets, and camelids (llamas/alpacas) .

ANTIBIOTICS THAT KILL RABBITS

The narrow-spectrum, gram-positive oral antibiotics destroy the cecal microflora and cause fatal clostridial (*C. spiroforme*) enterotoxemia. Memorize with "**PLACE**":

Penicillins (ampicillin, amoxicillin) · **Lincomycin** · **Ampicillin** · **Clindamycin/Cephalosporins** · **Erythromycin**.

Same list applies to guinea pigs, chinchillas, and hamsters. Safe choices: **fluoroquinolones (enrofloxacin)**, **trimethoprim-sulfa**, **chloramphenicol**, and **metronidazole**; **parenteral** penicillin is acceptable in rabbits for treponematosis.

RAPID-FIRE RABBIT RECALL

Head tilt + posterior paresis + white cataract → ***E. cuniculi*** → **fenbendazole**, shed in **urine**, **zoonotic**.

Crusty perineum and face lesions → **rabbit syphilis** → **parenteral penicillin**.

Crusted ear canal → ***Psoroptes cuniculi*** → **selamectin**.

Peracute death + liver necrosis + bleeding → **RHDV (calicivirus)** — **reportable**.

Fever/acute death in Oregon/California, cottontail source → **myxomatosis** → **PCR**.

Nasal discharge → **snuffles (*Pasteurella*)** → **nasal culture**.

Ringworm → **terbinafine**.

7 | Ferrets

7.1 Endocrine & Neoplastic Disease

Condition	Presentation	Diagnosis & Treatment
Adrenal disease ("ferret hyperadrenocorticism")	The main endocrinopathy of the ferret; onset around 3.5 years of age. Symmetric truncal alopecia starting at the tail base, pruritus, vulvar swelling in spayed females , prostatic enlargement and stranguria in males, and return of sexual behavior.	Abdominal ultrasound to find the enlarged adrenal, plus a sex-steroid (adrenal) panel . Treat by adrenalectomy or deslorelin implant.

WHY FERRET ADRENAL DISEASE IS NOT CLASSIC CUSHING'S

Despite the name, ferret adrenal disease is **not hyperadrenocorticism**. The tumors secrete **SEX STEROIDS** — **estradiol, androstenedione, 17-OH-progesterone** — **NOT cortisol**.

That is why an **ACTH stimulation test and LDDS are unhelpful**, and why the diagnostic test is a **sex-steroid panel plus abdominal ultrasound**. The pathogenesis is linked to **early neutering** and loss of gonadal negative feedback on LH.

Condition	Presentation	Key Point
Hyperestrogenism / estrogen toxicity	Caused by persistent estrus — the jill is an induced ovulator , so an unbred, unmated female stays in estrus and estradiol rises unchecked.	Leads to bone-marrow suppression → APLASTIC ANEMIA (pancytopenia). Prevent by spaying, hCG/GnRH, or a vasectomized hob.
Splenomegaly	Extremely common on physical exam.	Usually an INCIDENTAL finding (extramedullary hematopoiesis). Do not chase it unless there are other abnormalities.

7.2 Infectious Disease

Disease	Agent	Presentation	Management
Canine Distemper	Morbillivirus (paramyxovirus)	Essentially 100% fatal in ferrets — the virus both ferrets and dogs get. Classic signs: pruritic chin/inguinal rash, hyperkeratosis of the footpads, ocular-nasal discharge, then CNS disease.	Vaccinate (ferret-approved recombinant vaccine). No treatment; euthanasia.
Aleutian Disease	Parvovirus	Weight loss + hind limb weakness + hypergammaglobulinemia + glomerulonephritis.	The tested concept is the mechanism: disease is caused by immune-complex deposition , not direct viral cytopathic effect — which is what distinguishes it from every other parvoviral disease you have learned.
Epizootic Catarrhal Enteritis ("green slime disease")	Coronavirus	Vomiting and diarrhea — profuse bright-green mucoid stool. Older ferrets often become chronic.	Supportive care; introduction of a new ferret is the classic history.
Influenza 	Influenza A	Ferrets are bidirectionally zoonotic with humans — they catch human flu and give it back. Sneezing, fever, nasal discharge.	Amantadine has been used; supportive care is the mainstay in practice. Owners with flu should avoid handling.

7.3 Urinary & Reproductive

Topic	Key Facts
Urolithiasis	CYSTINE uroliths are the predominant ferret stone on modern diets; struvite was the historical stone, seen on plant-protein-based diets with alkaline urine. Male ferrets obstruct readily — a urethral obstruction is a surgical emergency.
Induced ovulator	Ferrets join rabbits, cats, and camelids on the induced-ovulation list. This is the root of hyperestrogenism.

RAPID-FIRE FERRET RECALL

Tail-base alopecia + vulvar swelling in a 3.5-year-old → **adrenal disease (sex steroids, not cortisol)**.

Pancytopenia in an intact jill in prolonged estrus → **hyperestrogenism** → **aplastic anemia**.

Weight loss + hind limb weakness + hypergammaglobulinemia → **Aleutian disease (immune complexes)**.

Bright green diarrhea after a new ferret arrived → **ECE (coronavirus)**.

Big spleen, otherwise well → **incidental**.

100% mortality virus shared with dogs → **canine distemper**.

8 | Guinea Pigs

Topic	Key Facts
Hypovitaminosis C (scurvy) <i>the single most-tested guinea pig fact</i>	Guinea pigs (like primates and humans) lack L-gulonolactone oxidase , the enzyme converting L-gulonolactone → L-ascorbic acid , so vitamin C is a dietary essential. Deficiency → defective collagen: lameness and reluctance to move, swollen painful joints, gingival bleeding, poor wound healing, and rough coat. Supplement with fresh vegetables and stabilized vitamin C pellets. Vitamin C degrades rapidly in stored pellets and in water.
Vitamin C excess	Excess vitamin C predisposes to CALCIUM OXALATE UROLITHIASIS (ascorbate is metabolized to oxalate). Guinea pigs are already prone to urolithiasis; do not oversupplement.
Contraindicated antibiotics	Penicillins (ampicillin, amoxicillin), cephalosporins, clindamycin, lincomycin, and erythromycin. Cause fatal clostridial dysbiosis. The same list applies broadly to rabbits, chinchillas, and hamsters.
Pneumonia	Two agents: <i>Bordetella bronchiseptica</i> and <i>Streptococcus pneumoniae</i> . Separately, <i>Streptococcus equi</i> subsp. zooepidemicus causes cervical lymphadenitis ("lumps") and can extend to pneumonia. Rabbits are asymptomatic <i>Bordetella</i> carriers — never house them with guinea pigs.
Coccidiosis	Predisposed in the YOUNG. <i>Eimeria caviae</i> ; diarrhea in weanlings; treat with sulfonamides and improve sanitation.

RAPID-FIRE GUINEA PIG RECALL

Won't walk, painful joints, bleeding gums → **vitamin C deficiency.**

Bladder stone → **calcium oxalate**, worsened by **too much** vitamin C.

Cervical "lumps" → ***Strep equi* subsp. zooepidemicus.**

Housed with a rabbit, now has pneumonia → ***Bordetella bronchiseptica.***

Any penicillin → **dead guinea pig.**

9 | Chinchillas

Topic	Key Facts
Fatal antibiotic dysbiosis	Amoxicillin (and the whole penicillin/cephalosporin/clindamycin/lincomycin/erythromycin group) causes fatal dysbiosis. Use FLUOROQUINOLONES instead (enrofloxacin), or trimethoprim-sulfa, chloramphenicol, or metronidazole.
Bacterial enteritis	Three organisms causing diarrhea/gastroenteritis: <i>Proteus</i>, <i>E. coli</i>, and <i>Pseudomonas</i> — all GRAM NEGATIVE .
<i>Pseudomonas</i> infection	Causes conjunctivitis with a white discharge (and otitis, metritis, septicemia). Waterborne — contaminated water bottles are the classic source. Treat with aminoglycosides — gentamicin or neomycin.
Dermatophytosis Z	<i>Trichophyton mentagrophytes</i> dominates in chinchillas and small exotics — it does NOT fluoresce under a Wood's lamp. Add antifungal powder to the dust bath.

DERMATOPHYTE SPECIES BY HOST — A TABLE WORTH MEMORIZING WHOLE

Dogs/cats: *Microsporum canis* (most common; **fluoresces apple-green**), *M. gypseum* (geophilic, does **not** fluoresce), *T. mentagrophytes* (less common, zoonotic, does **not** fluoresce).

Chinchillas / rabbits / small exotics: *T. mentagrophytes* dominates → treat with **terbinafine**.

Cattle: *T. verrucosum* — thick gray-white crusts, especially **around the eyes/face in calves**.

Poultry: *Trichophyton (Lophophyton) gallinae* = **favus** / "**white comb**" → miconazole.

Remember: **only *M. canis* reliably fluoresces**, and only about half the time. Culture (DTM) is definitive.

10 | Hamsters, Rats, Mice & Hedgehogs

10.1 Hamsters

Condition	Key Facts
Renal amyloidosis	Older Syrian (golden) hamster with PU/PD, weight loss, and ascites. Leads to glomerulonephropathy and chronic renal failure with severe proteinuria and hypoalbuminemia (hence the ascites). More common in females. Diagnose with CONGO RED stain (apple-green birefringence under polarized light).
"Wet tail" (proliferative ileitis)	<i>Lawsonia intracellularis</i> — the same organism as porcine proliferative enteropathy. Watery diarrhea, wet perineum, high mortality in recently weaned/shipped 3–8 week old hamsters. Treat with tetracycline/metronidazole and fluids; stress reduction is key.
Lymphocytic choriomeningitis (LCM) 	Arenavirus. Hamsters, mice, and rats are the reservoir hosts. A type IV hypersensitivity disease — the immune response, not the virus, causes the lesions. Zoonotic ; teratogenic in human pregnancy.

10.2 Rats & Mice

Condition	Key Facts
Chromodacryorrhea ("red tears")	Porphyrin secretion from the Harderian gland behind the eye in rats. It is NOT blood — it is a non-specific stress indicator . Look for the underlying husbandry problem or disease.
Tail ring / ringtail	Annular constrictions of the tail from low environmental humidity . Prevent by maintaining humidity at 40–70% ; treat by adding fatty acids to the diet and lanolin to the tails .
Mousepox	Ectromelia virus (an orthopoxvirus) — lethal ; causes limb necrosis and amputation ("ectromelia" = missing limb). A major laboratory-colony catastrophe.
Epizootic diarrhea of infant mice (EDIM)	Rotavirus. Yellow, watery diarrhea in suckling mice; low mortality but high morbidity.
Mouse hepatitis virus (MHV)	Coronavirus. One of the most prevalent and disruptive infections of laboratory mouse colonies.
Lice	<i>Polyplax spinulosa</i> — the SUCKING louse of rats (narrow, pointed head). Rats do NOT get chewing lice . <i>P. serrata</i> is the mouse equivalent.
Tropical rat mite 	<i>Ornithonyssus bacoti</i> — a mite from rats that will infest a person's home and bite humans (papular dermatitis). Treat with pyrethroids or fipronil PLUS environmental decontamination — treating only the rat fails.

10.3 Hedgehogs

Condition	Key Facts
Wobbly Hedgehog Syndrome	A progressive demyelinating disease causing ataxia and ascending paralysis in African pygmy hedgehogs, typically beginning before 2 years of age. Starts with hind-limb ataxia and progresses to tetraparesis over months. Suspected hereditary; no treatment; diagnosis is definitive only on post-mortem histopathology.

RAPID-FIRE RODENT & HEDGEHOG RECALL

Old Syrian hamster, PU/PD, ascites → **renal amyloidosis** → **Congo red**.

Wet tail → ***Lawsonia intracellularis***.

Red tears in a rat → **chromodacryorrhea** — **porphyrin, not blood, means stress**.

Constricted dry tail → **ringtail** — **raise humidity to 40–70%**.

Mite from rats infesting the owner's house → ***Ornithonyssus bacoti***.

Sucking louse on a rat → ***Polyplax spinulosa***.

Lethal mouse poxvirus → **ectromelia**.

Hamster/mouse arenavirus, zoonotic, type IV HSR → **LCMV**.

Ataxic young hedgehog → **wobbly hedgehog syndrome** — **demyelinating, no treatment**.

11 | Nonhuman Primates

PRIMATE MEDICINE IS ZOONOSIS MEDICINE

Essentially every primate question on the NAVLE is really a **zoonotic risk / occupational safety** question. Know the direction of transmission for each agent.

Agent	Family	Disease	Direction & Significance
Rubella (German measles)	Togavirus	Mild; not a concern in nonhuman primates.	Contrast with rubeola below — this is the "not the answer" option.
Rubeola (measles)	Paramyxoviridae — morbillivirus	Serious, fatal disease in nonhuman primates: rash, respiratory signs, and conjunctivitis .	Transmitted from HUMANS to PRIMATES — anthroponosis. Vaccinate staff and quarantine.
Marburg virus Z	Filoviridae — same family as Ebola ; enveloped, negative-sense RNA	Viral hemorrhagic fever. Named for Marburg, Germany , where it was identified in 1967 after outbreaks in laboratory workers handling African green (vervet) monkeys imported from Uganda .	BSL-4 pathogen. Natural reservoir: fruit bats.
Herpes B (<i>Cercopithecine herpesvirus 1</i> / <i>Macacine alphaherpesvirus 1</i>) Z	Alphaherpesvirus	Nearly asymptomatic in the macaque host (oral vesicles). In humans, a MACAQUE BITE or scratch causes fatal ascending encephalomyelitis .	The single most important occupational hazard in primate medicine. Immediate scrub and post-exposure acyclovir/valacyclovir.
Hepatitis A Z	Picornavirus	Affects Old World primates (chimpanzees and macaques) and causes hepatitis.	Fecal–oral transmission; zoonotic .
Hepatitis B	Hepadnavirus	Affects chimpanzees , which become chronic carriers .	Blood-borne; the chronic carrier state is the tested point.

RAPID-FIRE PRIMATE RECALL

Macaque bite → **Herpes B** → **fatal human encephalitis**.

Rash + respiratory + conjunctivitis, fatal, from a human caretaker → **rubeola (morbillivirus)**, not rubella.

Filovirus from African green monkeys, 1967, Germany, fruit bat reservoir → **Marburg (BSL-4)**.

Fecal–oral zoonotic hepatitis of Old World primates → **Hep A**. Chronic carrier chimps → **Hep B**.

Primates, like guinea pigs, **cannot synthesize vitamin C**.

Appendix A | Reportable & Zoonotic Quick Sheet

HOW REPORTABLES ARE TESTED

The NAVLE rarely asks you to treat a reportable disease. It asks **what you do next**. The answer is almost always: **notify the State Animal Health Official / USDA Area Veterinarian in Charge, do not move animals, and depopulate as directed**. If a question offers "treat with X" alongside "contact the authorities" for one of the diseases below, pick the authorities.

Reportable / Foreign Animal Diseases in This Guide

Species Group	Disease	Agent	The Correct Action
Fish	Tilapia Lake Virus	Orthomyxovirus	Notify authorities; depopulate. 10–90% mortality.
	Infectious Salmon Anemia	Orthomyxovirus (isavirus)	No treatment; eradicate the entire stock .
	Infectious Hematopoietic Necrosis	Novirhabdovirus	Notify; endemic PNW/Alaska.
	Megalocytivirus (ISKNV / scale drop)	Iridovirus	PCR of spleen & kidney , then report.
	Epizootic Ulcerative Syndrome	<i>Aphanomyces invadans</i>	Contact the federal authorities .
Avian	Highly Pathogenic Avian Influenza 	Influenza A	RT-PCR of choanal cleft & tracheal swabs ; immediate report; depopulate.
	Virulent Newcastle Disease	Avian paramyxovirus-1	Report; proventricular/gizzard petechiae.
	Infectious Laryngotracheitis	Herpesvirus	Report; bloody beaks and gasping.
	Infectious Bursal Disease	Birnavirus	Report; the reportable immune-system disease of birds.
	Avian Chlamydiosis / Psittacosis 	<i>Chlamydia psittaci</i>	Report to public health; doxycycline 45 days.
Rabbit	Salmonella pullorum / gallinarum	<i>Salmonella</i>	NPIP test-and-slaughter.
	Rabbit Hemorrhagic Disease (RHDV-2)	Calicivirus	Foreign animal disease — report; liver necrosis + bleeding.
	Myxomatosis	Myxoma poxvirus	PCR; cottontail reservoir; CA/OR/Baja.

Zoonoses in This Guide

Agent	Animal Source	Human Disease & Transmission
<i>Chlamydia psittaci</i>	All birds	Psittacosis — atypical pneumonia. Inhalation of aerosolized dried droppings.
Avian influenza	Poultry, waterfowl	Conjunctivitis to fatal pneumonia. Occupational.
<i>Mycobacterium avium</i>	Birds, reptiles	Disseminated disease in immunocompromised people.
<i>Cryptococcus neoformans</i>	Pigeon droppings	Meningitis in immunocompromised people; birds are environmental hosts, not clinically ill.
<i>Encephalitozoon cuniculi</i>	Rabbits — shed in URINE	Microsporidiosis; beware the immunosuppressed owner .
Herpes B virus	Macaques — bites/scratches	Fatal ascending encephalomyelitis . The great occupational hazard of primate work.

Agent	Animal Source	Human Disease & Transmission
Marburg virus	African green monkeys; fruit bat reservoir	Viral hemorrhagic fever. BSL-4.
Hepatitis A	Old World primates	Fecal–oral hepatitis.
LCMV (arenavirus)	Hamsters, mice, rats	Aseptic meningitis; teratogenic in pregnancy. Type IV hypersensitivity pathology in the host.
<i>Ornithonyssus bacoti</i>	Rats	Tropical rat mite — infests homes , causes papular dermatitis in people.
Dermatophytes (<i>T. mentagrophytes</i>)	Rabbits, chinchillas, small exotics	Ringworm. Does not fluoresce.
Pentastomes	Reptiles	Visceral pentastomiasis; humans are accidental intermediate hosts.
<i>Salmonella</i>	Reptiles, amphibians, poultry, backyard flocks	The classic pet-reptile and backyard-chicken zoonosis. Advise hand hygiene and no reptiles in homes with children <5.
Influenza A	Ferrets — bidirectional	Ferrets catch human flu and transmit it back.
<i>Gnathostoma</i>	Fish	Humans acquire it by ingesting undercooked fish → migratory cutaneous swellings and, rarely, CNS disease. The fish-borne nematode zoonosis to name.
Tularemia	Rabbits and rodents (via ticks)	Dogs and cats become exposed by hunting tick-bitten rabbits and rodents ; then infect people.

Appendix B | Therapeutics & Antibiotic Safety by Species

B.1 The Hindgut-Fermenter Antibiotic Rule

APPLIES TO RABBITS, GUINEA PIGS, CHINCHILLAS, AND HAMSTERS

NEVER give orally: Penicillins (ampicillin, amoxicillin) · Cephalosporins · Clindamycin · Lincomycin · Erythromycin.

Why: these narrow-spectrum, gram-positive-targeting drugs wipe out the normal cecal flora, allowing *Clostridium* (*C. spiroforme*, *C. difficile*) overgrowth → fatal enterotoxemia.

SAFE: fluoroquinolones (enrofloxacin) · trimethoprim-sulfa · chloramphenicol · metronidazole · aminoglycosides (gentamicin, neomycin) · tetracyclines.

The exception: parenteral penicillin G is the treatment of choice for **rabbit syphilis**. The route, not the drug, is what kills.

B.2 Drug-to-Disease Quick Table

Drug	Species	Indication
MS-222 (tricaine)	Fish	Anesthesia and euthanasia. 21-day withdrawal . Buffer with bicarbonate.
Aquacalm (metomidate)	Ornamental fish	Sedation/anesthesia — ornamental (non-food) fish only.
Formalin / hydrogen peroxide	Fish	<i>Saprolegnia</i> and other water molds.
Copper sulfate / H ₂ O ₂ / chloroquine	Marine fish	<i>Amyloodinium</i> (velvet disease).
Doxycycline ± tobramycin	All birds	<i>Chlamydia psittaci</i> — 45-day course.
Itraconazole	Pet birds	Aspergillosis. (Avoid in African greys.)
Nystatin	Pet birds	Candidiasis (individual bird).
Copper sulfate	Poultry	Candidiasis (flock, in water).
Miconazole	Poultry	Favus / white comb.
Ivermectin	Birds, snakes, lizards	<i>Knemidocoptes</i> , snake mites. FATAL IN CHELONIANS.
Amprolium	Poultry	Coccidiosis (thiamine analog).
Fenbendazole	Poultry	<i>Ascaridia</i> , <i>Heterakis</i> , <i>Capillaria</i> .
Fenbendazole	Rabbits	<i>Encephalitozoon cuniculi</i> — 28 days.
Sulfonamides	Birds	Coccidiosis.
Sulfonamides + pyrimethamine	Birds	Leucocytozoonosis.
Chloroquine + primaquine	Penguins, raptors	Avian malaria (<i>Plasmodium</i>).
Tetracycline	Poultry	Fowl cholera; <i>Bordetella avium</i> secondaries.
Enrofloxacin	Birds (individual)	Severe bumblefoot (systemic). Prohibited extralabel in poultry.
Selamectin	Rabbits	<i>Psoroptes cuniculi</i> ear mites.
Terbinafine	Rabbits, chinchillas	Ringworm (<i>T. mentagrophytes</i>).
Penicillin G — PARENTERAL only	Rabbits	Rabbit syphilis (<i>Treponema paraluis-cuniculi</i>).
Gentamicin / neomycin	Chinchillas	<i>Pseudomonas</i> .
Fluoroquinolones	Chinchillas, rabbits, GPs	The default safe antibiotic class.
Metronidazole	Snakes	<i>Entamoeba invadens</i> .

Drug	Species	Indication
Amantadine	Ferrets	Influenza (per source; supportive care in practice).
Pyrethroids / fipronil	Rats	<i>Ornithonyssus bacoti</i> — plus environmental decontamination .
Vitamin C	Guinea pigs	Scurvy. Do not oversupplement → calcium oxalate stones.
Riboflavin	Chicks	Curled toe paralysis — fully reversible if caught early .